

Anurag Sanjay Ghosh

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WORK EXPERIENCE

Ribbon Communications, Westford, MA

January 2025 – Present

Data Scientist 3

- Built an **AI-driven log analytics platform** that processes **millions** of unstructured log lines, transforming them into structured, correlated event streams for anomaly detection and root cause analysis, reducing investigation latency by **~35–40%**.
- Engineered a **log-mining pipeline** combining template learning, entity resolution, **BERTopic-based** clustering, & temporal correlation modeling, reducing log volume by over **50%** & improving signal-to-noise ratio in event detection.
- Secured a purchase order from **Altice** for the log analytics platform, enabling enterprise adoption of Ribbon's Analytics platform.
- Developed machine learning models for optical network performance monitoring using in-band signal power features, applying **Quantile XGBoost and Gaussian Process regression** to predict **Optical Signal-to-Noise Ratio (OSNR)**.
- Leveraged **physics-informed** feature engineering to mitigate data sparsity challenges, improving predictive performance by **~2×**.
- Built a conversational analytics tool for automated exploratory data analysis, enabling natural language-driven dashboard creation in Apache Superset using multi-agent **LlamaIndex** workflows with **fine-tuned** SQL agent & tool-augmented execution.

Data Science Intern

- Designed a **website categorization** service to catalog & capture network traffic trends with a **RoBERTa** model trained on **1M** webpages scraped by **Selenium** crawlers. Extended with **zero-shot** classification (**DeBERTa**) for fine-grained subcategorization.
- Delivered an F1 score of **85%** & reduced unknown/new URL cataloging & processing time by **70%** on the client side (**AT&T**).

Universal Music Group, New York City, NY

July 2023 – December 2023

Data Science Analyst Intern

- Trained **LightGBM-based Customer LTV Models** on **Sagemaker** for predicting purchase propensity & identifying superfans (high-value customers). Processed **4 million online transactions** to train the model, & interpreted predictions using **Shapley**.
- Elevated leadership's awareness of a **data-driven** approach to fanbase analysis & customer valuation, leading to its prioritization.
- Prepared a **feature extraction module** for promotional emails to categorize them & identify engagement-driving features.
- Devised a **BERT-based active-learning method** with unsupervised clustering to predict email categories & recommend key linguistic & structural attributes contributing to high customer engagement rates.
- Secured a **10% increase** in the performance of downstream **in-production** engagement assessment models.

WINES Lab, Northeastern University, Boston, MA

January 2023 – September 2023

Machine Learning Research Assistant

- Explored machine learning-based **radio fingerprinting** by training 2 deep learning models, **AlexNet & Vision Transformers**, on sequences of in-phase (I) and quadrature (Q) data from **LoRa** devices to identify the devices in varying external scenarios.
- Attained **99%** classification accuracy across all device configurations under the same-day training and testing conditions.

TIAA GBS, Mumbai, Maharashtra

July 2019 - July 2022

Analyst, Software Developer

- Led a team of 4 developers overseeing the **development & production** of **multiple** Spring Boot microservice applications.
- **Independently** supported & collaborated with the business on live deployments for **retirement income evaluation apps**.
- Oversaw **20** successful **production releases** with **OpenShift** integration, reducing deployment time from hours to a few minutes.

ACADEMIC PROJECTS

Bone Fracture Detection and Localization

February 2023 - April 2023

- Trained a **DETR & Vision Transformers-based** pipeline for bone fracture localization & classification from X-Ray images.
- Conducted a comparative study between transformer-based and CNN-based approaches to evaluate performance and robustness.
- Achieved **mAP of 0.78**, **precision of 0.82**, and **recall of 0.76**, outperforming traditional RCNN and YOLO based pipelines

PUBLICATIONS

"Hybrid Image Encryption Technique Using Genetic Algorithm and Lorenz Chaotic System" ITM Web of Conferences 32, 03009 (2020), https://www.itm-conferences.org/articles/itmconf/pdf/2020/02/itmconf_icacc2020_03009.pdf

EDUCATION

Northeastern University, Khoury College of Computer Sciences, Boston, MA

December 2024

Master of Science, Artificial Intelligence

4.00 GPA

Relevant Coursework: Large Language Models, Reinforcement Learning, NLP, Algorithms, Machine Learning.

Teaching Assistant: Foundation of Artificial Intelligence (CS5100)

Mumbai University, Mumbai, India

June 2019

Bachelor of Engineering, Computer Engineering

TECHNICAL SKILLS

Programming/Scripting: Python, JavaScript, Java, C++, C.

Databases and Cloud Service: Google Cloud Platform, Big Query, AWS, Sagemaker. Azure, MySQL, Microsoft SQL Server

Tools & Frameworks: pandas, Docker, Python (Keras, PyTorch, TensorFlow, Pandas, Scikit-learn, Streamlit, Llamaindex, LangGraph